

Food Delivery Company System

Object-Oriented Programming Project

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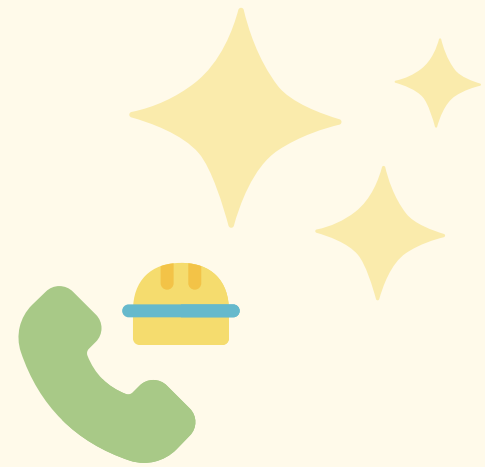
24-0768 Dana Abu Quta

24-0747 Lama Al Maharibi

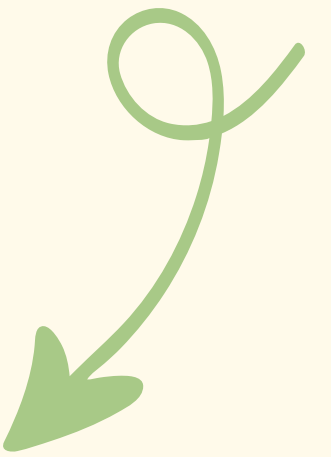
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System Overview



The system has three main users:

Customer

- Registers and logs in
- Creates orders
- Adds items to cart
- Proceeds to checkout
- Views current order status

Restaurant

- Registers and logs in
- Creates and updates menu
- Views assigned orders
- Updates order status when food is ready

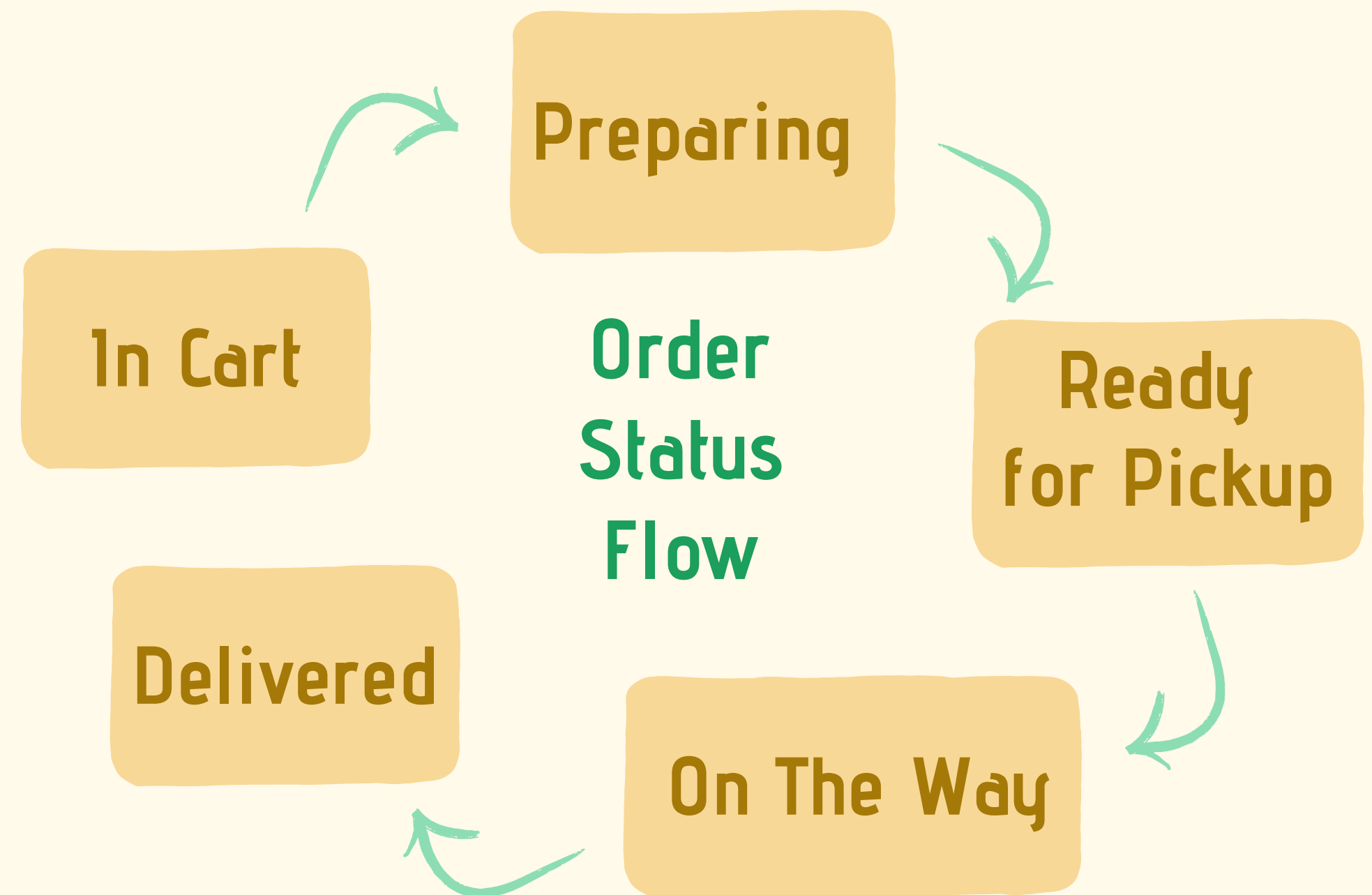
Driver

- Registers as Bike Driver or Car Driver
- Views assigned order
- Updates delivery status
- Tracks earnings

Main System Features and Flow

Main System Features

- User registration and login
- Restaurant menu management
- Cart and order creation
- Driver assignment
- Cash and card payment
- Delivery fee and total price calculation
- Order status tracking
- Driver earnings update



System Controller

Company

- Controls the main system flow
- Manages customers, restaurants, drivers, orders, checkout, and login

Driver Classes

Driver

- Parent class for drivers

BikeDriver / CarDriver

- Different driver types with different delivery fee and time calculations

Code Structure

User and Order Classes

Customer

- Stores customer information
- Used when creating and viewing orders

Order

- Stores order details, items, status, delivery fee, and payment
- Connects customer, restaurant, driver, and payment

Restaurant Classes

Restaurant

- Stores restaurant information
- Manages assigned orders

Menu

- Stores item names and prices
- Allows adding, editing, displaying, and removing items

Payment Classes

Driver

- Parent class for payments

CashPayment / CardPayment

- Different payment types with different receipt details



Inheritance and Polymorphism

Driver

Payment

Inheritance

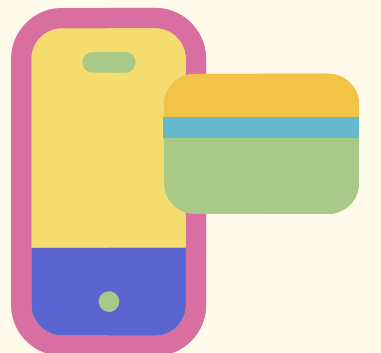
- Driver is the parent class
- BikeDriver inherits from Driver
- CarDriver inherits from Driver

- Payment is the parent class
- CashPayment inherits from Payment
- CardPayment inherits from Payment

Polymorphism

- calcDeliveryFee() is overridden in BikeDriver and CarDriver
- calcEstTime() is overridden in BikeDriver and CarDriver
- Each driver type calculates delivery fee and time differently

- paymentReciept() is overridden
- getPaymentType() is overridden



Other OOP Concepts Used

Encapsulation

- Private and protected attributes
- Public setters and getters

Constructors and Destructors

- Constructors collect required information when objects are created
- Destructors clean dynamically allocated objects

Operator Overloading

- `operator<<` is overloaded to display order details

Composition

- Restaurant has a Menu
- This shows a HAS-A relationship

Pointers

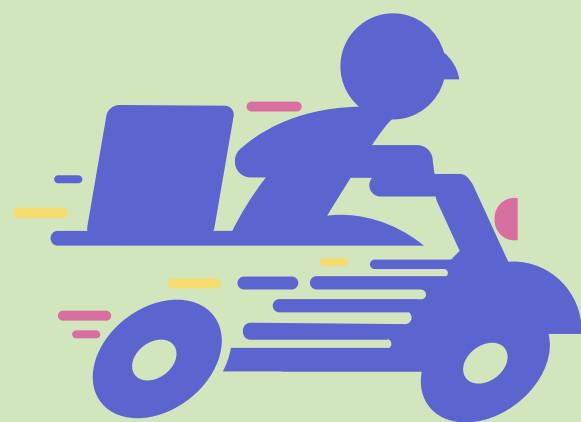
- Used to connect related objects
- Example: Order has pointers to Customer, Restaurant, Driver, and Payment

Exception Handling

- Used to handle invalid selections

Example:

- Invalid restaurant number
- Invalid menu item
- Invalid item removal



Thank
you